B25 FIT Level 01 Project
Smart Garden Management System

Name	Index Number	Degree Program
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INTRODUCTION

- Our project is a **Smart Garden Management System** integrated with **Artificial Intelligence (AI)**, designed for real-world applicability in modern agriculture. It consists of **five interdependent subsystems**, all centrally controlled by a **Raspberry Pi**.
- The system collects real-time data through sensors and uses AI-based decision making to automatically manage plant care. Unlike ordinary smart gardens, this system features **advanced automation, subsystem interdependency, and built-in security features**, making it a more intelligent and reliable solution.

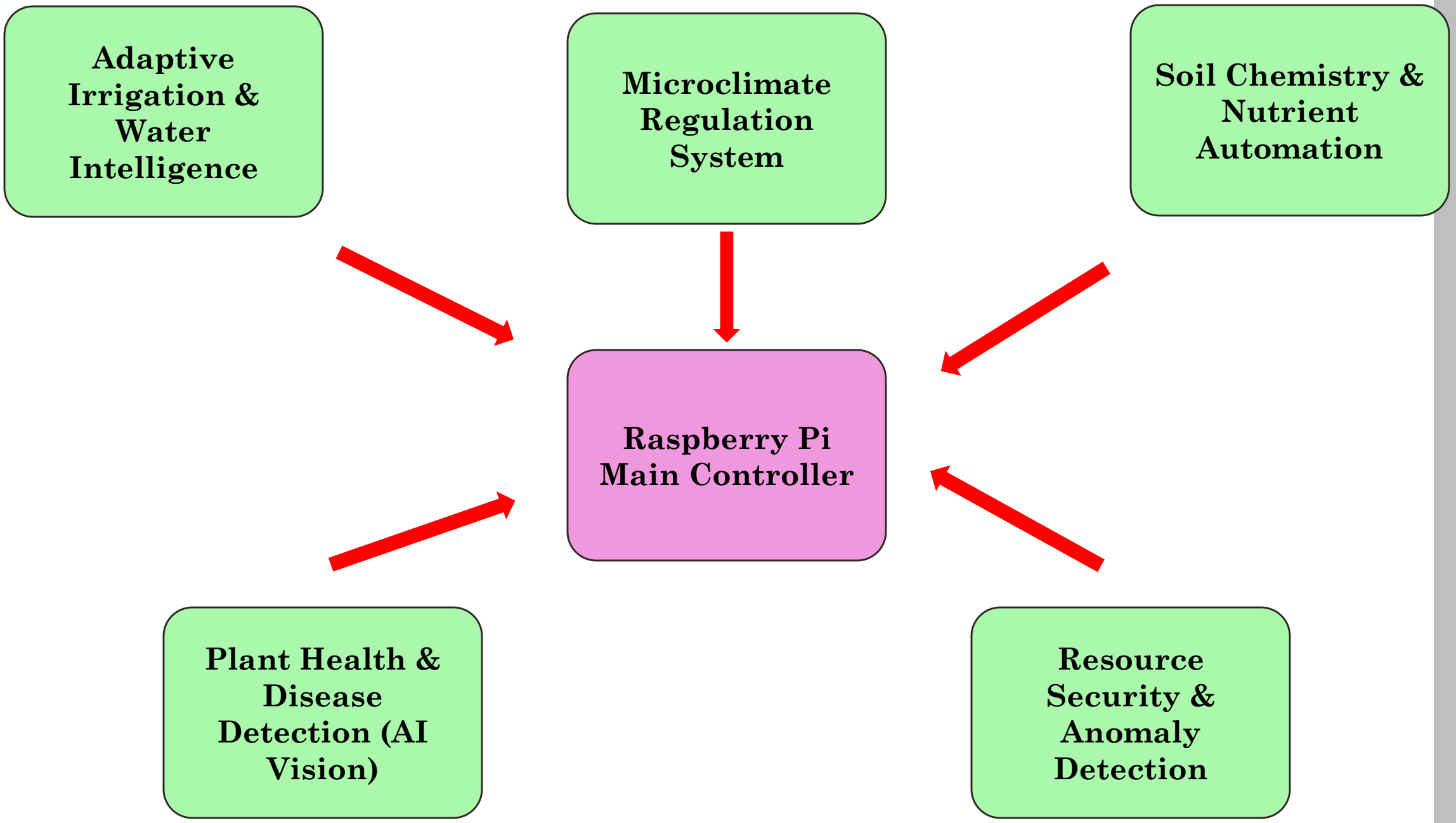
Problems & Solutions

- **Problem**
- In industrial agriculture, traditional and basic smart gardening methods rely on manual monitoring or isolated systems, leading to inefficient resource usage, poor environmental control, and delayed detection of critical issues.
- **Proposed Solution**

We propose an **industry-oriented Smart Garden Management System** integrated with **AI** and controlled by a **Raspberry Pi**, consisting of **five interdependent subsystems**. The system uses real-time sensor data and centralized AI-based decision-making to automate operations, optimize resource usage, and enhance reliability with built-in security features.

Sub Systems

- Adaptive Irrigation & Water Intelligence
- Microclimate Regulation System
- Soil Chemistry & Nutrient Automation
- Plant Health & Disease Detection (AI Vision)
- Resource Security & Anomaly Detection


Adaptive Irrigation & Water Intelligence
Microclimate Regulation System
Soil Chemistry & Nutrient Automation
Raspberry Pi Main Controller
Plant Health & Disease Detection (AI Vision)
Resource Security & Anomaly Detection

01.Adaptive Irrigation & Water Intelligence

- **1. Multi-Parameter Decision Engine**

Water Need = f(moisture + temperature + humidity + time + plant type)

- **2. Predictive Irrigation**

Weather API(rain prediction)+Evapotranspiration estimation

- **3. Flow Intelligence & Leak Detection**

Water flow sensor + pressure sensor

- **4. Multi-Zone Smart Irrigation** – Zone1 + Zone2 + Zone3+ Zone4
- **5. Water Optimization Al** - Minimize water usage
- **6. Learning Capability** – Store past data

Input

- **Soil Moisture Sensors (Zones)**
- **Humidity Sensors**
- **Temp Sensors**
- **Flow Sensor**
- **Time of day**

Process

- **ESP 32 (wi-Fi)**
- **Raspberry Pi (AI engine)**
- **Water Need Process(**Weather API(rain prediction)+Evapotranspiration estimation)

Output

- **Solenoid Valves (Zones)**
- **Water Pump**

02.Microclimate Regulation System

- 1. Multi-Parameter Climate Intelligence

= (temperature + humidity + CO_2 + plant profile)

2. Dynamic Ventilation Control -
High temp + high humidity → strong ventilation

3. Intelligent Humidity & Mist Control

Dry + hot = mist ON , Humid already = NO mist

4. CO_2 Optimization System - High CO_2 → increase airflow

5. Time-Based Climate Profiling – Day and Night Profiles

Input

- **Temp Sensor**
- **Humidity Sensor**
- **CO_2 Sensor**

Process

- **ESP 32 (wi-Fi)**
- **Raspberry Pi (AI engine)**
- Ventilation and mist process by temperature + humidity + CO_2 + plant profile

Output

- **Fans**
- **Mist Systems**
- **Vent Motors**

03.Soil Chemistry & Nutrient Automation

- 1. Multi-Parameter Soil Intelligence

PH, EC (Electrical Conductivity), Soil moisture

- 2. Nutrient Balance Decision Engine

Nutrient Requirement = f(pH + EC + crop type + growth stage)

- 3. Automated Nutrient Dosing System
- 4. Crop-Specific Nutrient Profile

System adapts automatically based on selected crop

- 5. Fault & Safety System

pH not changing after dosing → sensor fault

Input

- pH Sensor
- EC Sensor
- Moisture Sensor

Process

- ESP 32 (wi-Fi)
- Raspberry Pi (AI engine)
- Processing Nutrient requirement via pH and EC sensors

Output

- Nutrient Pump
- pH Up Pump
- pH Down Pump

04.Plant Health & Disease Detection (AI Vision)

- 1. Generalized Plant Health Detection (Works for ALL Plants)
- Universal plant health indicators→ Leaf color changes
- 2. AI Model Strategy
- Healthy, Diseased, Nutrient deficient
- 3. Color & Pattern Analysis
- High green → healthy, Yellow edges → nutrient issue
- 4. Growth Monitoring
- 5. Integration with Other Subsystems
- Detect yellow leaves → signal nutrient system
- ESP 32 cam mount in a path planning robot

Input

- Camera Module (ESP32-CAM)
- Path for the robot to move
- Ultrasonic sensor
- Wheel Encoder
- IMU (MPU6050)

Process

- ESP 32 (wi-Fi)
- Raspberry Pi (AI engine)
- Health Analysis
- Stores map and Path planning
- Process data, wich are getting from the ESP 32 cam

Output

- Action Commands to: (Irrigation / Nutrient System)
- Gear motors

05.Resource Security & Theft Detection

- 1. Resource Monitoring
- Water tank level, Water usage rate, Power consumption
- 2. Physical Security System
- Detect human/animal movement
- 3. Asset Protection Logic
- Detect plant removal (weight change)
- 4. Smart Alert System
- 5. Integration with Other Subsystems
- Low tank → notify irrigation subsystem

Input

- Water Level Sensor
- Flow Sensor
- Current Sensor
- Load Sensor
- ESP32-CAM

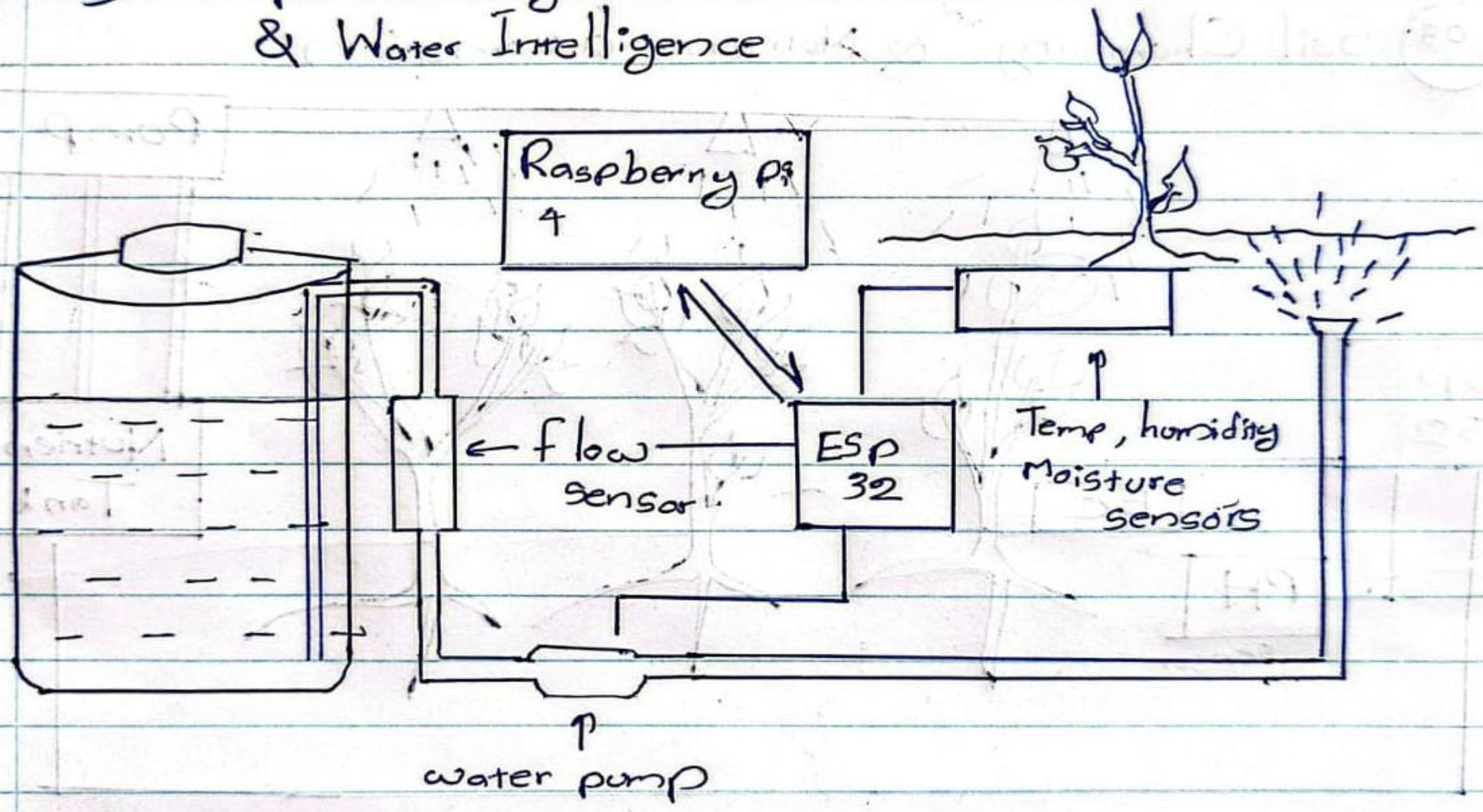
Process

- ESP 32 (wi-Fi)
- Raspberry Pi (AI engine)
- Anomaly Detection & Theft Alerts

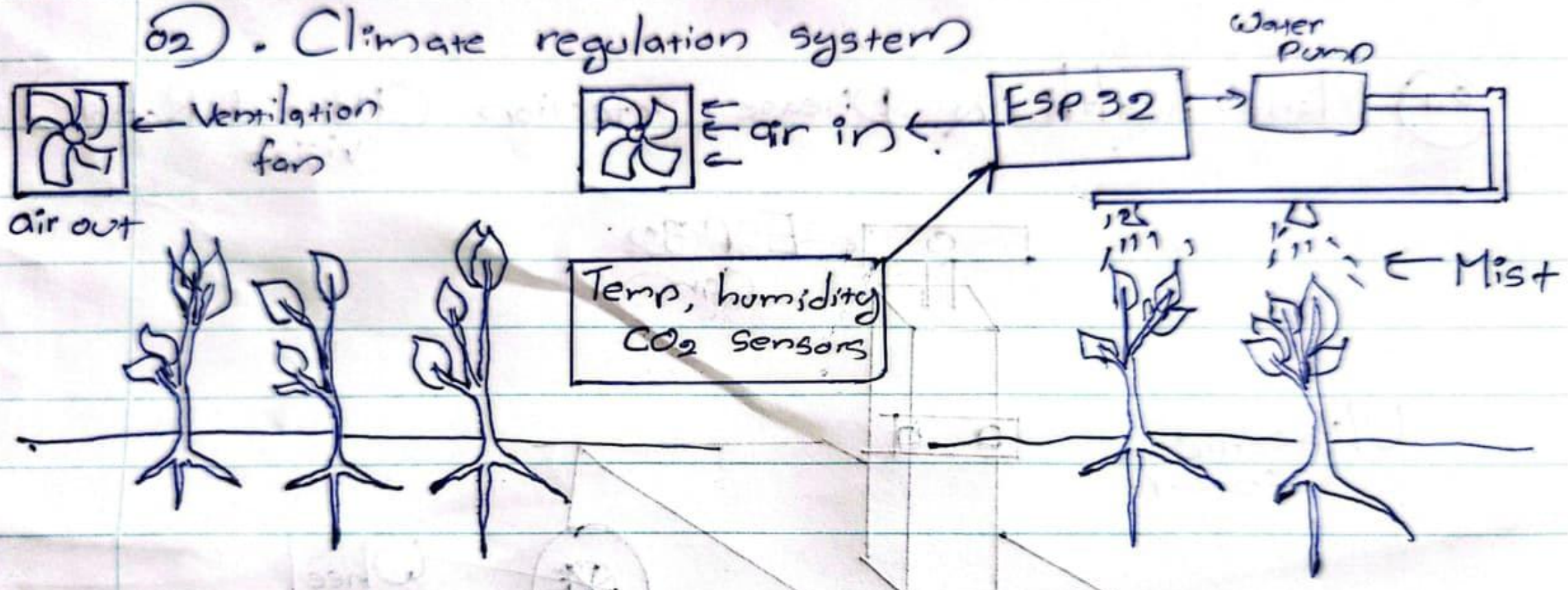
Output

- Stop Systems (Pump/Valves)
- Send Alerts

01). Adaptive Irrigation
& Water Intelligence



02). Climate regulation system



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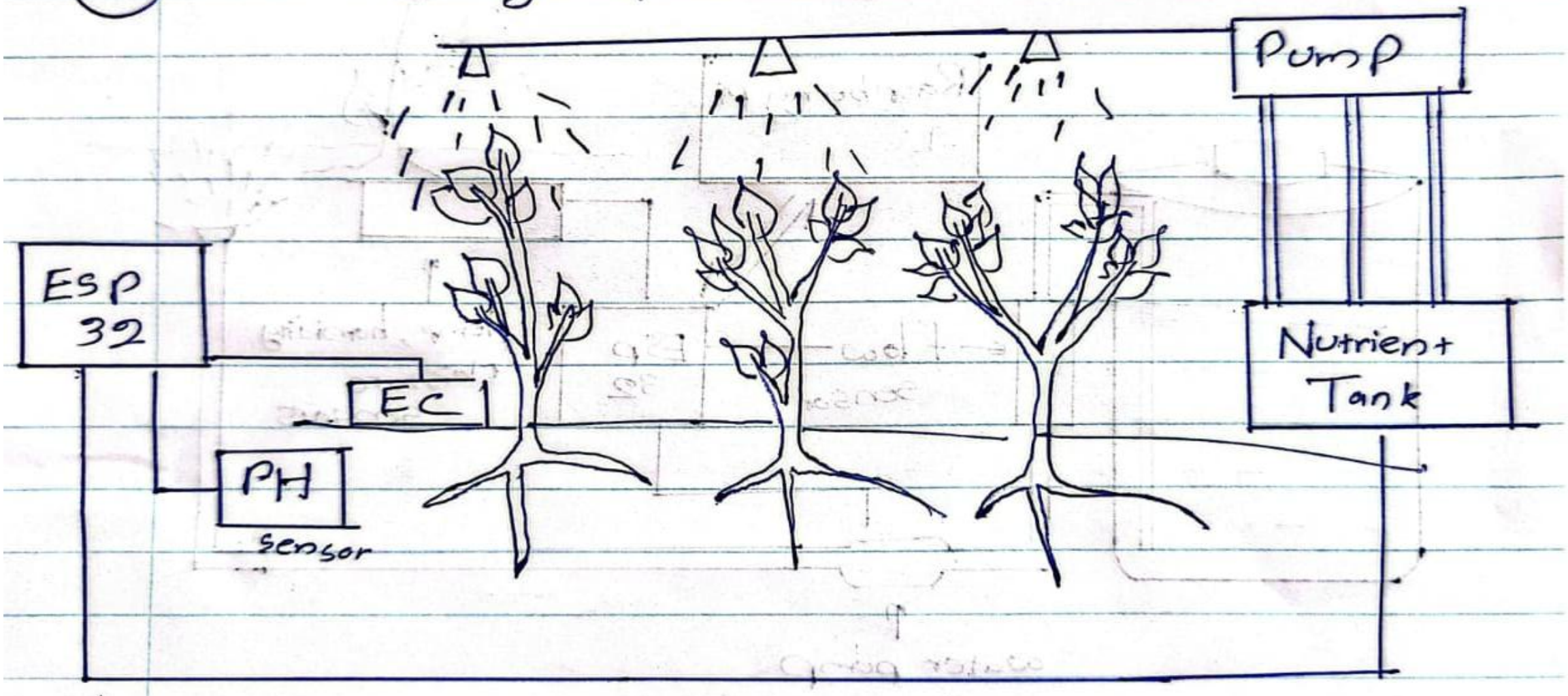
03 Soil Chemistry & Nutrient automation



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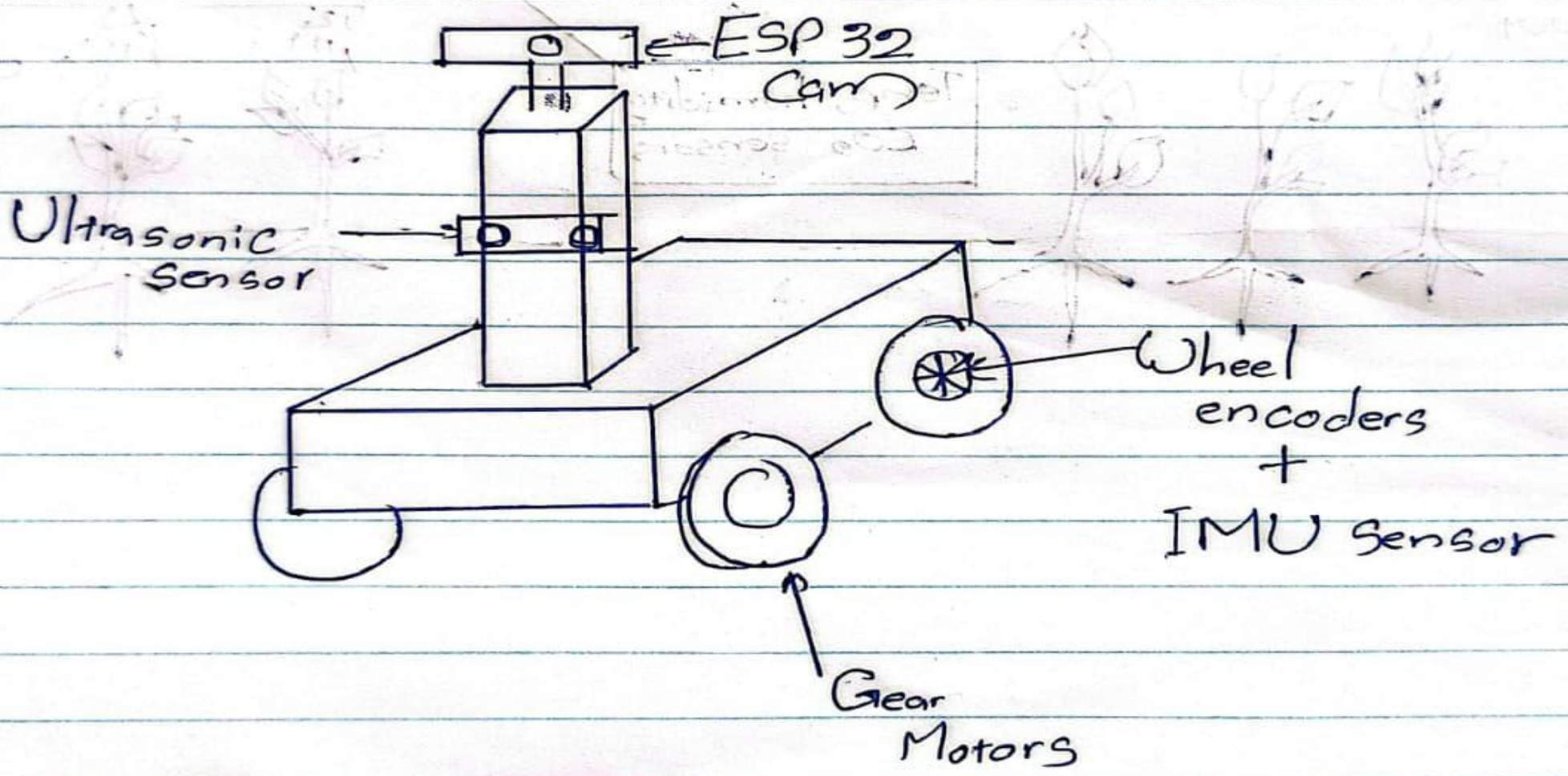
(04) Plant health & Disease Detection (AI Vision + Robot)



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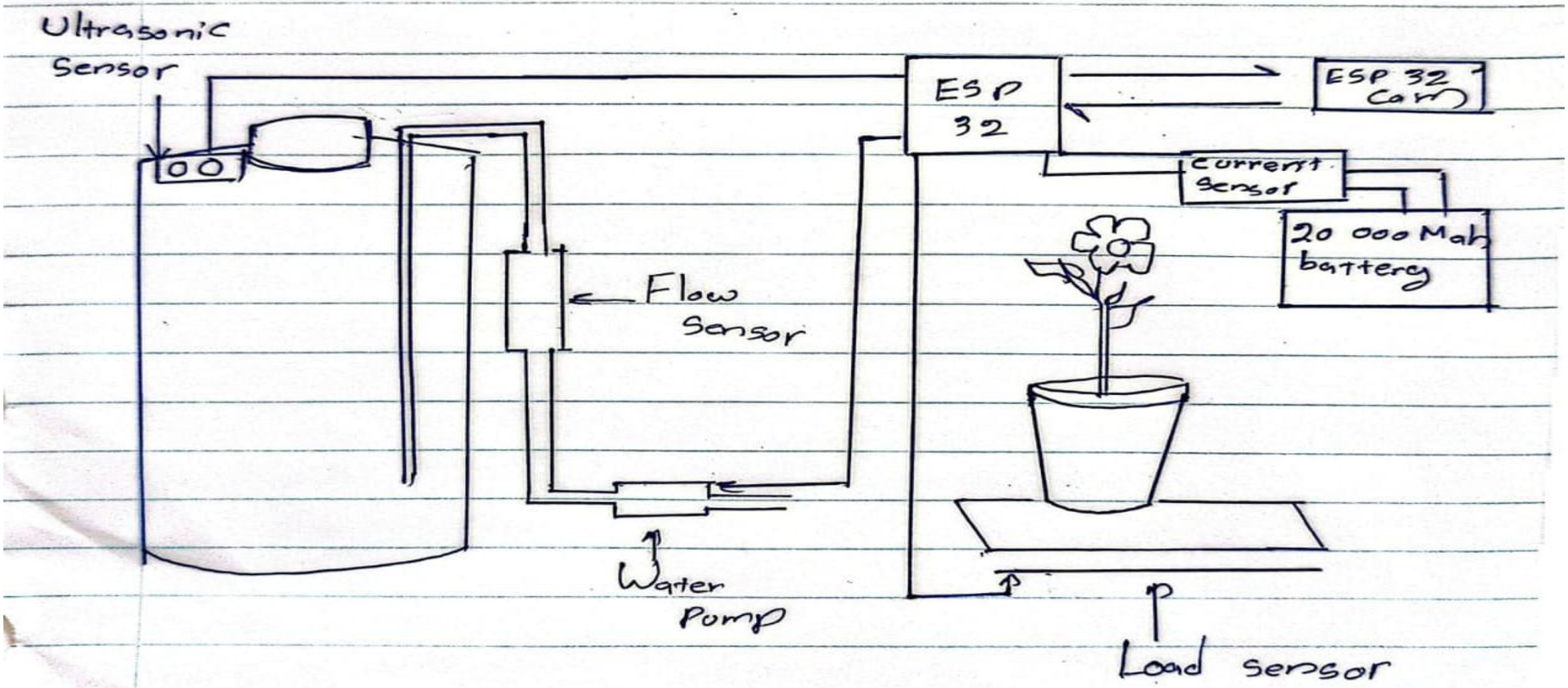
(5) Resource security & Theft detection



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SMART GARDEN MANAGEMENT SYSTEM
(5 SUBSYSTEMS INTEGRATED WITH RASPBERRY PI)

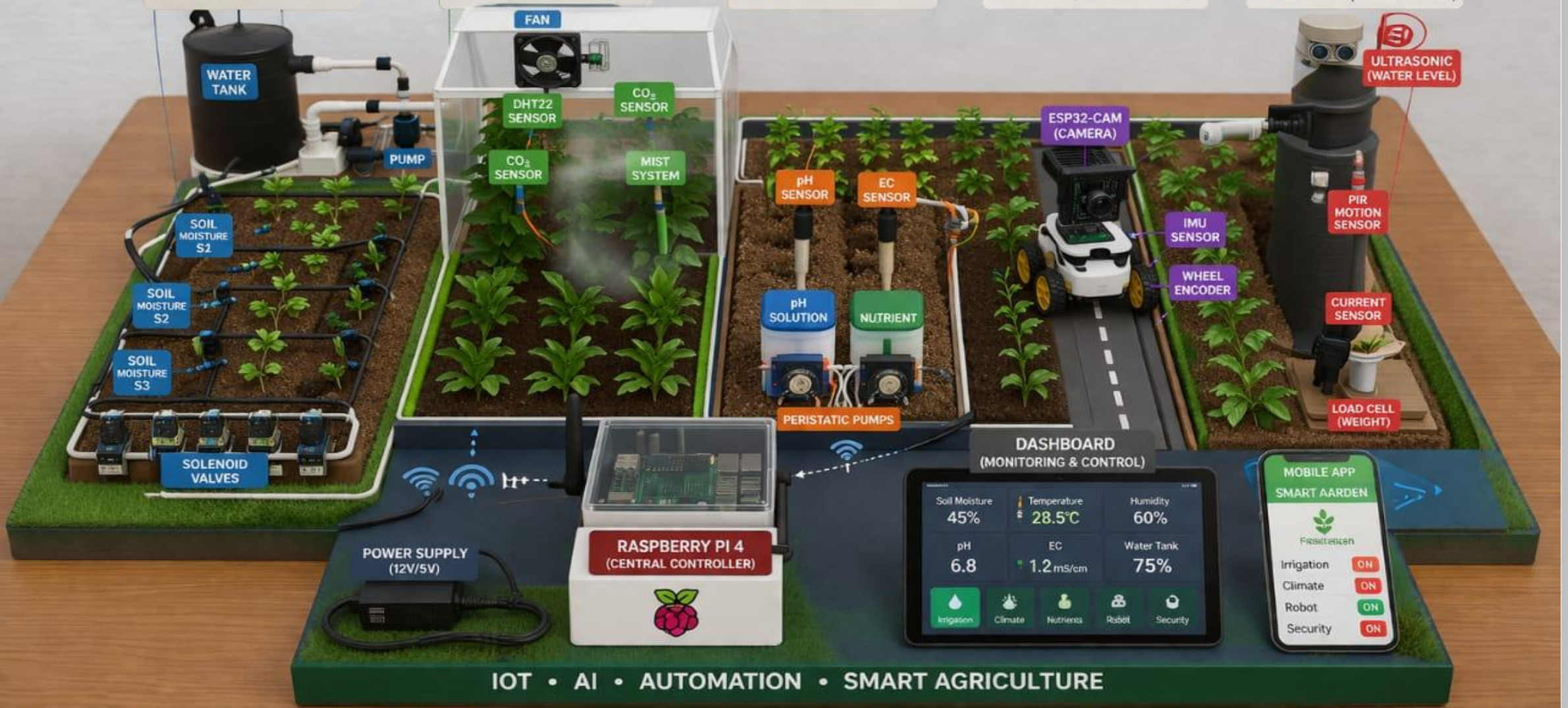
1 ADAPTIVE IRRIGATION & WATER INTELLIGENCE
- ESP32
- Soil Moisture Sensors
- Flow Sensor
- Solenoid Valves
- Water Pump

2 MICROCLIMATE REGULATION SYSTEM
- ESP32
- Temperature & Humidity (DHT22) Sensor
- CO_2 Sensor
- Fans / Vent / Mist System

3 SOIL CHEMISTRY & NUTRIENT AUTOMATION
- ESP32
- pH Sensor
- EC Sensor
- Peristaltic Pumps
- Nutrient Tanks

4 PLANT HEALTH & DISEASE DETECTION (AI VISION ROBOT)
- ESP32-CAM Camera
- ESP32 (Robot Control)
- IMU Sensor
- Wheel Encoders
- Motors (Robot Movement)

5 RESOURCE SECURITY & ANOMALY DETECTION
- ESP32
- Ultrasonic Sensor (Water Level)
- Current Sensor
- PIR Motion Sensor
- Load Cell (Theft Detection)



IOT • AI • AUTOMATION • SMART AGRICULTURE

Input

- Soil Moisture
- Flow Sensor
- Temp Sensor
- Humid Sensor
- CO_2 Sensor
- pH Sensor
- EC Sensor
- Camera Module
- IMU + Encoder
- Ultrasonic
- Current Sensor
- Load
- ESP 32 cam

Process

- WiFi Communication (MQTT)
- Raspberry Pi (Main System)
- AI Decision Engine
- Data Processing
- Central Control

Output

- Solenoid Valves
- Water Pump
- Fans/Mist
- Nutrient Pumps
- Alerts
- Motor driver

Components

Components	Usage
Raspberry Pi 4	Main controller for processing data, and managing communication between all components
ESP 32	Getting Inputs from each sub system , wireless connection between Raspberry Pi and Control actuators
ESP 32 Camera	Takes images of plant leaves for analysis
Motor Driver	To control gear motors
Gear motors	To move the chassis
Soil Moisture Sensors	TO measure

Component	Usage
pH Sensor	Measures soil acidity/alkalinity to ensure proper nutrient absorption.
EC Sensor	Measures nutrient concentration in soil to manage fertilizer levels.
Ultrasonic Sensor	Measures water tank level and helps monitor resource availability.
Current sensor	**(ACS712)** Monitors power usage of devices (pumps, motors) to detect faults or abnormal behavior.
Load sensor	Detects weight changes to identify plant removal or theft.
HX711 Module	Converting analog signal into digital
12 volt supply	Power supply
Temp & Humidity Sensors	Monitors environmental conditions for climate control.
Flow Sensor	Measures water flow rate to control irrigation and detect leaks or blockages.

Solenoid Valves (Zones)	Automatically control water flow to different irrigation zones.
Water pump	Water feeding
CO_2 Sensor	Measures carbon dioxide levels to optimize plant growth and ventilation.
Fans	Ventilation
Mist System	Using Sprinklers to generate mist
Wheel Encoder	Measures wheel rotation to calculate distance traveled and speed of the robot. It helps in tracking position and ensuring accurate movement along the planned path.
IMU (MPU6050)	**(Inertial Measurement Unit)** Measures orientation, acceleration, and rotation (using accelerometer + gyroscope). It helps the robot maintain direction, detect turns, and improve navigation accuracy.

Expected Budget

Component	Price	Quantity	Total
Raspberry Pi 4	24000.00	1	24000.00
ESP32	1000.00	5	5000.00
ESP 32 Cam	1500.00	1	1500.
Temp & Humidity sensor	1000.00	2	2000.00
Motor Driver	1000.00	1	1000.00
Gear motors	1000.00	4	4000.00
Soil Moisture Sensors	300.00	1	300.00
Solenoid Valves (Zones)	2000.00	3	6000.00
Water pump	1200.00	4	4800.00
CO_2 Sensor	3000.00	1	3000.00
Fans	2000.00	2	4000.00
Mist System	2500.00	2	5000.00
Wheel Encoder	500.00	2	1000.00
IMU (MPU6050)	750.00	1	750.00

pH Sensor	2500.00	2	5000.00
EC Sensor	3000.00	1	3000.00
Ultrasonic Sensor	800.00	1	800.00
Current sensor	1000.00	1	1000.00
Load sensor	1500	1	1500.00
HX711 Module		1	
12 volt supply	2000.00	1	2000.00
Flow sensor	1000.00	1	1000.00
Water Pipes	1500.00		1500.00
Robot chassis	2000.00		2000.00
Wires	1000.00		1000.00
Alarms	1200.00	1	1200.00
Total			**82500.00**

Technologies and Softwares

- **Communication Technology → WiFi Communication-MQTT Protocol**
- **Arduino IDE** – Programming ESP32 and ESP32-CAM
- **Python** – Main system programming on Raspberry Pi
- **Node-RED / Flask (optional)** – Dashboard & control interface
- **TensorFlow Lite** – Lightweight AI model for plant health detection
- **OpenCV** – Image processing (color, pattern analysis)

index		Hardware task	Software task	Study task
254226E		ESP32 , Soil moisture sensors (multi-zone), Flow sensor, Solenoid valves, Water pump	Arduino IDE, (ESP32 programming) , MQTT communication, Python (decision logic on Raspberry Pi)	Smart irrigation systems, Sensor calibration, Water flow analysis, Feedback control systems
254211E		ESP32 DHT22 (temperature & humidity sensor) CO_2 sensor Fans, vents, mist system	Arduino IDE, MQTT, Python (climate control logic)	Environmental control systems, Greenhouse climate management, Multi-variable decision making
255071F		ESP32 , pH sensor, EC sensor, Peristaltic pumps, Liquid nutrient tanks	Arduino IDE, MQTT, Python (nutrient calculation logic)	Soil chemistry (pH & nutrients), Fertigation systems, Closed-loop automation
254242A		ESP32-CAMESP32 (robot control), Motors + motor driver, Wheel encoders, IMU sensor	Arduino IDE, Python (Raspberry Pi)Open, CV (image processing), TensorFlow Lite (AI model)	Computer vision, Machine learning basics, Robotics & path planning, Sensor fusion (IMU + encoder)
254044T		ESP32, Ultrasonic sensor, (water level), Current	Arduino IDE, MQTT,	Anomaly detection

References

- https://opencv.org/
- https://www.arduino.cc/
- https://flask.palletsprojects.com/
- https://www.tensorflow.org/
- https://Daraz.lk/

- **THANK YOU !**